

System Design & Architecture Document (SDD)

SmartFarm Management Application

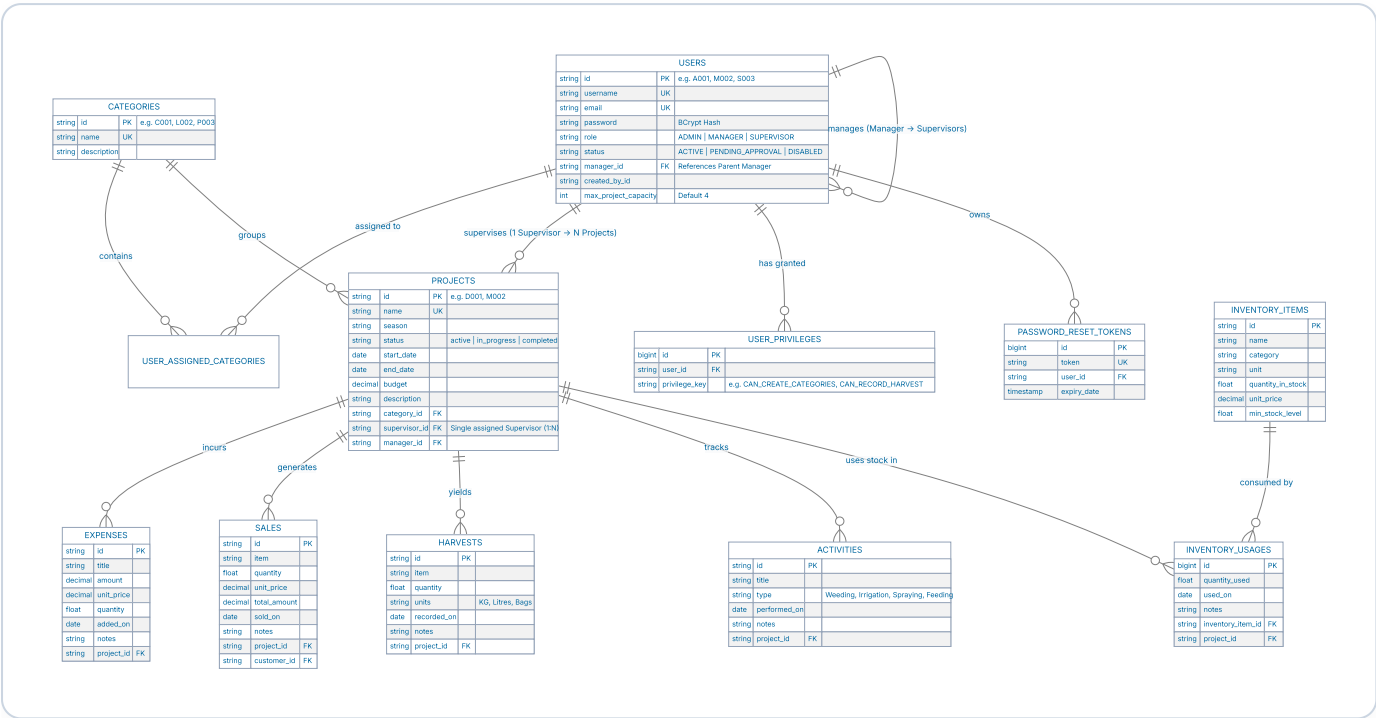
Document Version: 1.0	SDLC Stage: Phase 2 — System Architecture & Design	Status: Approved Technical Blueprint
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1. Architectural Overview & Technology Stack

The SmartFarm system follows a **3-Tier Clean Architecture** separating presentation, business orchestration, and persistent storage:

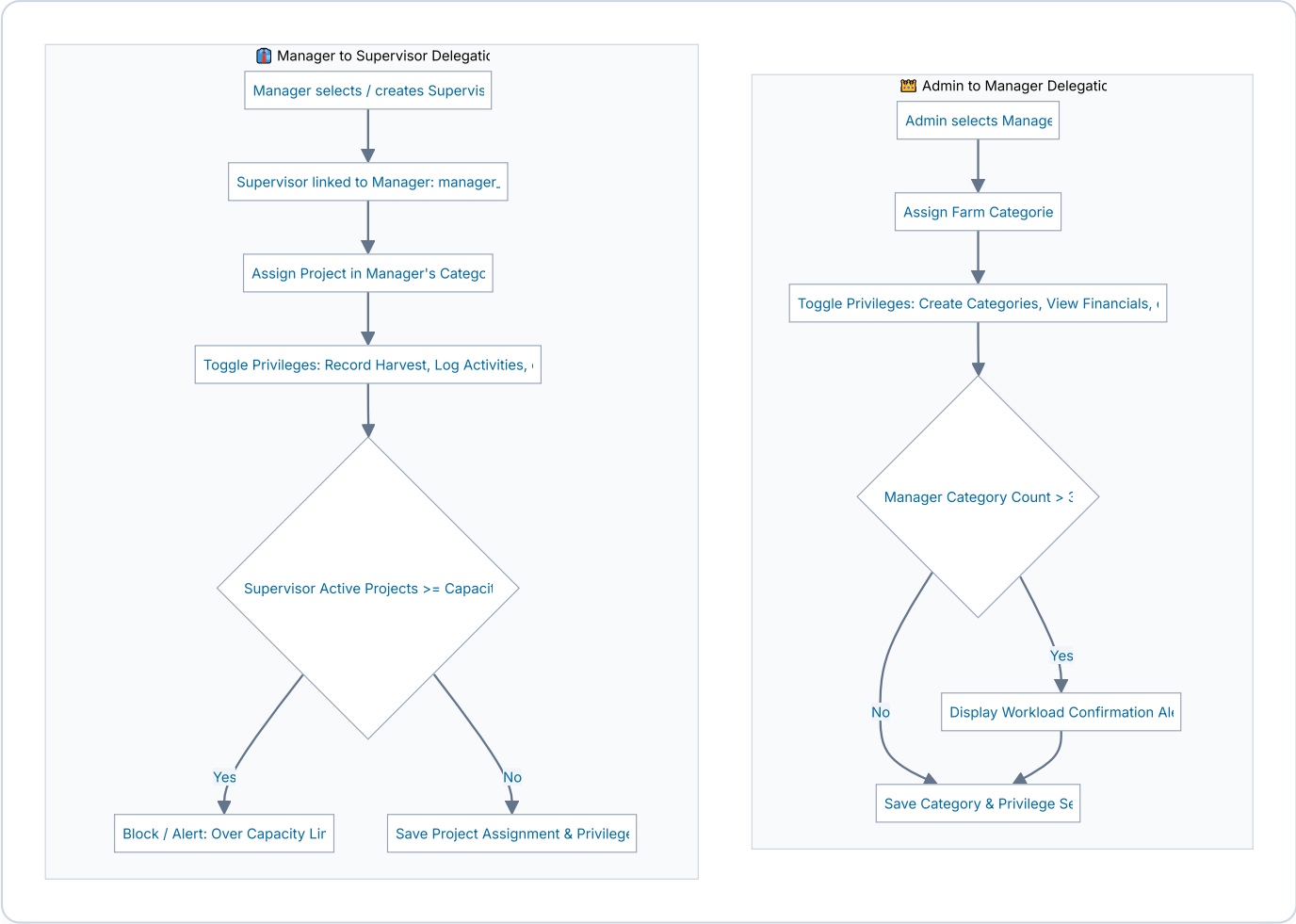


2. Complete Entity-Relationship Model (ERD)



3. Access Control & Privilege Delegation Engine (PBAC)

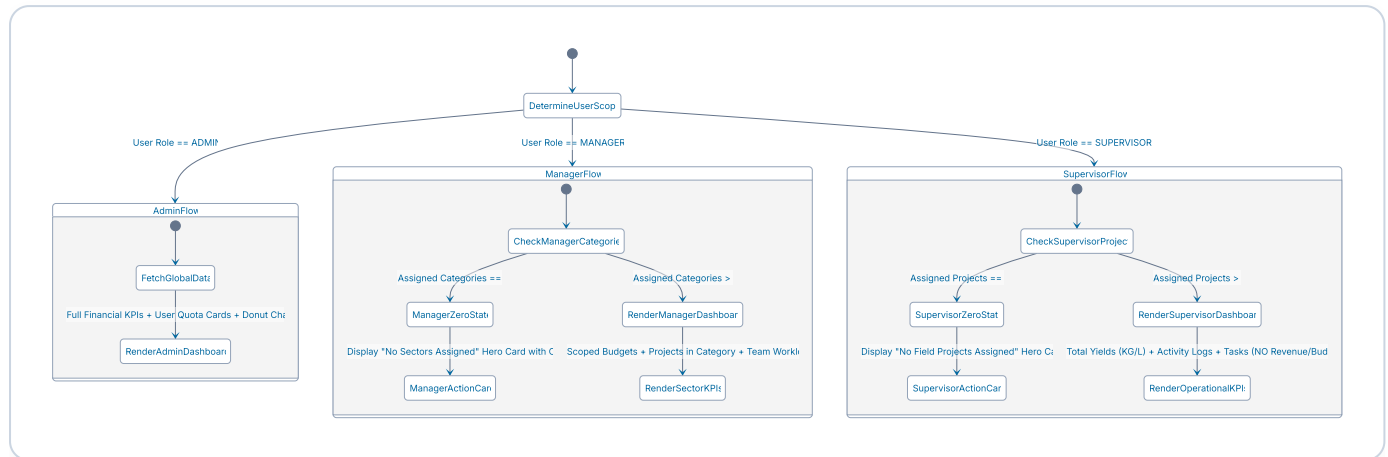
3.1 Delegation Hierarchy & Workload Guard



3.2 Privilege Definition Catalog

Privilege Key	Applied To	Backend Endpoint Guard	Default State
`CAN_CREATE_CATEGORIES`	`MANAGER`	`POST /categories`	Disabled (`false`)
`CAN_CREATE_SUPERVISORS`	`MANAGER`	`POST /users/staff` (Role = `SUPERVISOR`)	Enabled (`true`)
`CAN_VIEW_FINANCIALS`	`MANAGER`	`GET /projects/summary`, revenue fields	Enabled (`true`)
`CAN_MANAGE_BUDGETS`	`MANAGER`	`PUT /projects/{id}` (Budget property)	Disabled (`false`)
`CAN_RECORD_HARVEST`	`SUPERVISOR`	`POST /harvest`	Enabled (`true`)
`CAN_LOG_ACTIVITIES`	`SUPERVISOR`	`POST /activities`	Enabled (`true`)
`CAN_USE_INVENTORY`	`SUPERVISOR`	`POST /inventory/use`	Enabled (`true`)
`CAN_RECORD_EXPENSES`	`SUPERVISOR`	`POST /expenses`	Disabled (`false`)
`CAN_RECORD_SALES`	`SUPERVISOR`	`POST /sales`	Disabled (`false`)

4. Scoped Dashboard & Zero-State State Machine



5. API Specification & Security Contracts

5.1 Authentication & Password Recovery

POST /users/login : Authenticate via username/email + password; returns user payload with privileges and assigned categories.

POST /users/forgot-password : Validate email existence; if missing returns 400 with admin contact advice; if found sends reset email.

POST /users/reset-password : Consumes token and sets new password.

PATCH /users/{id}/admin-reset-password : Direct password reset by Admin or Manager.

5.2 Access Control & Delegation Endpoints

PATCH /users/{id}/categories : Admin assigns categories to Manager. Payload: `{ categoryIds: string[] }`.

PATCH /users/{id}/privileges : Assign fine-grained privilege set. Payload: `{ privileges: string[] }`.

PATCH /projects/{id}/supervisor : Manager assigns supervisor to project. Payload: `{ supervisorId: string }`.

GET /users/supervisors : Fetch supervisors managed by current manager.

5.3 Scoped Data & Financial Shielding Endpoints

GET /projects/summary : Scoped aggregate response:

- **Admin**: Global farm budget, active projects, all categories.

- **Manager**: Aggregates strictly for `assignedCategories`.

- **Supervisor**: Returns operational metrics (yield total, activity count); **omits budget/sales revenue**.

GET /categories :

- **Admin**: All categories.

- **Manager**: Only categories in `assignedCategories`.

- **Supervisor**: Only categories containing assigned projects.

6. Frontend Architecture & Zero-State Components

6.1 Component Hierarchy

```
App.jsx (Router)
├─ TopBar (Profile, Notifications, Dynamic Role Badges)
├─ MainLayout (Sidebar with Role-conditional Navigation)
└─ Pages
    ├─ DashboardPage (Dynamic Zero-State Engine & Scoped KPIs)
    ├─ ProjectListPage (Scoped Category Projects)
    ├─ ProjectFormPage (Date Validation startDate ≤ endDate)
    ├─ ProjectDashboardPage (Financial Shield for Supervisors)
    └─ UserManagementPage (Admin Quotas, Manager Team, Privilege Modals)
```

6.2 Zero-State Visual Specification

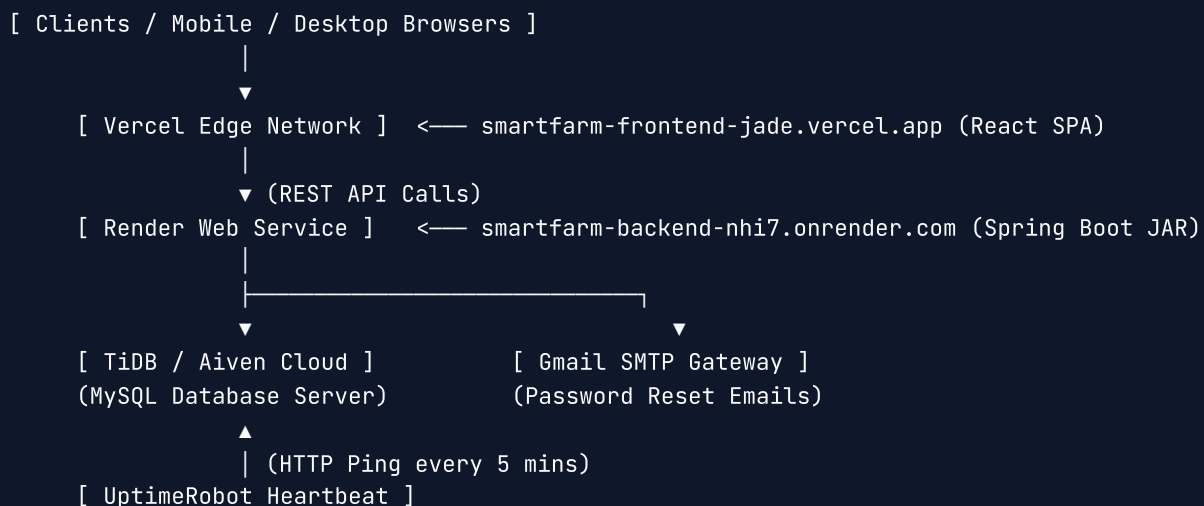
When an unassigned user visits `DashboardPage.jsx`:

Card Removal: Suppresses all stats grid cards, charts, and table rows.

Hero Message Card: Displays elevated card with icon, title, explanatory text, and action buttons based on granted privileges:

- *Manager without categories:*
- Text: `"You currently have no farm categories assigned to your management portfolio."`
- CTA 1: `[+ Create New Category]` (If `CAN_CREATE_CATEGORIES` is true).
- CTA 2: `[Contact Administrator]`.
- *Supervisor without projects:*
- Text: `"You have not been assigned to any active field projects yet."`
- CTA: `"Please notify your Farm Manager ([Manager Name]) to assign you a field project."`

7. Cloud Deployment Topology (\$0/month Architecture)



8. SDLC Traceability Matrix

Requirement ID (SRS)	Architectural Component (SDD)	Verification Method
FR-1.3 (Email Check on Forgot Password)	`UserService.forgotPassword`	Unit Test + Live Endpoint Test
FR-2.2 (Admin Manager PBAC Delegation)	`UserPrivilege`, `UserController`	MockMvc Test + Modal UI
FR-2.3 (Manager Supervisor PBAC)	`UserService.assignSupervisorPrivileges`	MockMvc Test + Modal UI
FR-3.2 (Supervisor Capacity Check)	`ProjectsService.assignSupervisor`	Capacity Validation Unit Test
FR-4.2 (Date Validation `start <= end`)	`ProjectsService` + `ProjectFormPage`	Form & API 400 Bad Request Test
FR-6.2 (Supervisor Financial Shield)	`ProjectDashboardPage.jsx` + API DTO	Supervisor Profile Assertion
FR-7.4 (Zero-State Dashboard)	`DashboardPage.jsx` Zero-State Engine	Browser Rendering Test
FR-8.2 (Safe Cascade Deletion)	`UserService.deleteUser`	Foreign Key Clean Cascade Test